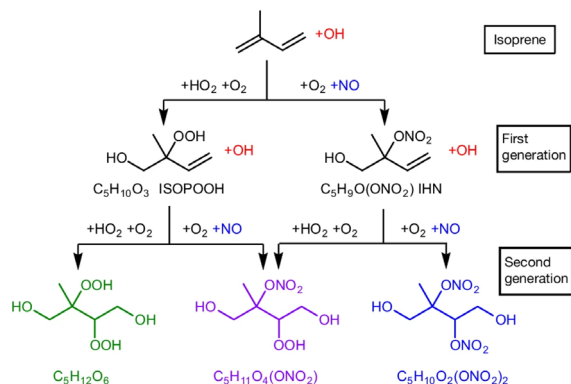


Isoprene Oxidation in the Atmosphere: From Classical Mechanisms to New Particle Formation

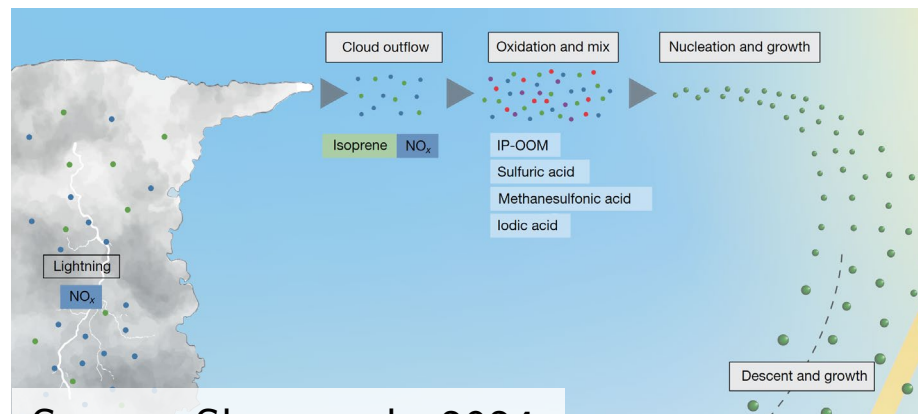
WONG, Hok Fong Silas (1155189917)

EESC3220: Atmospheric Chemistry

May 10, 2026



Source: Curtius et al., 2024



Source: Shen et al., 2024

Outline

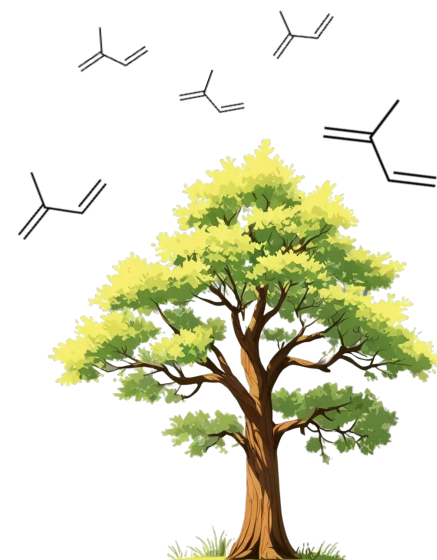
- 1 Introduction: Importance of Isoprene in the Atmosphere
- 2 Isoprene Oxidation Chemistry: Main Pathways and Mechanisms
- 3 New Particle Formation Dynamics

In-Situ Measurements of New Particle Formation in the Upper Troposphere

CERN CLOUD Experiment

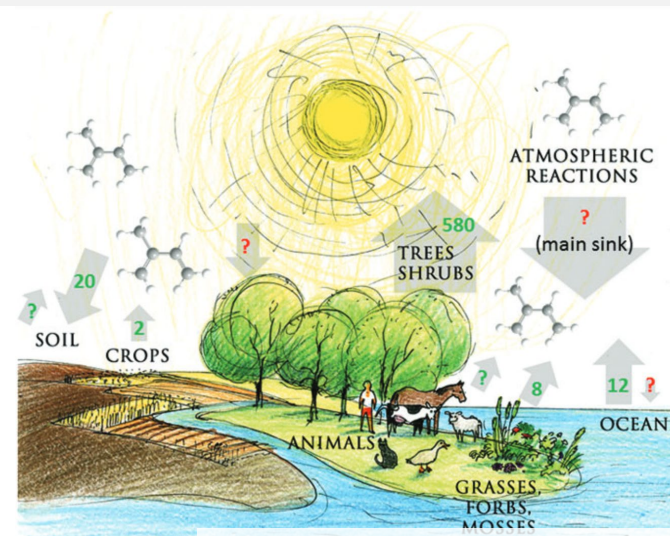
Diurnal Cycle of NPF

- 4 Implications and Conclusion



Importance of Isoprene in the Atmosphere

- Contributes to **half of the NMVOC emissions**, ~500 Tg/yr. Nearly the entire flux occurs during **daytime** from vegetation, especially broad-leaved trees (Russell et al., 2025; Wennberg et al., 2018).
- **Hotspots**: Amazon, Congo, SE Asian rainforests. Isoprene mixing ratio ranges between 1-20 ppbv in the continental boundary layer of the Amazon basin (Curtius et al., 2024).
- Highly reactive **diene**, oxidised by $\text{HO}\bullet$, O_3 , NO_3 to **form ozone and SOAs**.
- Limited aerosol formation at boundary-layer temperatures. First-generation oxidation products (ISOPOOH, IHN) have volatilities within the IVOC range. Isoprene must undergo **secondary oxidation, autoxidation or C_{10} accretion product formation** via RO_2 cross-reactions to contribute to particle growth.
- **High concentrations of <5 nm particles over the Amazon** unexplained for 20 years (Andreae et al., 2018; Curtius et al., 2024).



Source: Zhan et al., 2021

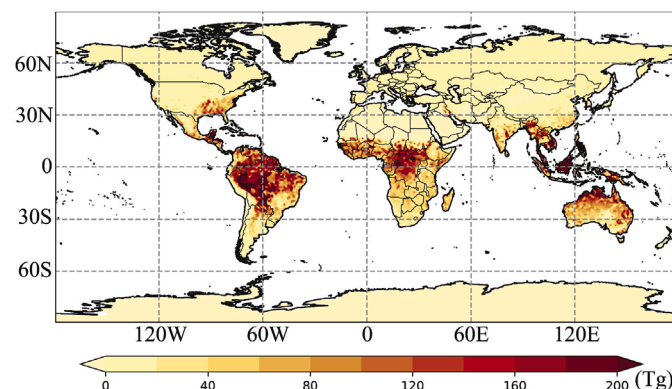


Figure. Long-term annual spatial distribution of isoprene emissions based on the CAMS inventory during the 2000-2019 period.

Source: Nyasulu et al., 2026

Main Destruction Pathways for Isoprene

- Under tropospheric conditions, alkanes mainly react with OH (daytime) and NO₃ radicals (nighttime).

Oxidant	$k_{\text{isoprene+oxidant}}$ (cm ³ molecule ⁻¹ s ⁻¹)
OH	$(2.7 \times 10^{-11}) \exp(390/T)$
O ₃	$(1.03 \times 10^{-14}) \exp(-1995/T)$
NO ₃	$(2.95 \times 10^{-12}) \exp(-450/T)$
Cl	$(7.6 \times 10^{-11}) \exp(-500/T)$

Source: IUPAC, 2006

Calculation. Typical [OH] ~ 1×10^6 molecules cm⁻³.

Upper tropospheric condition: 223-243 K,

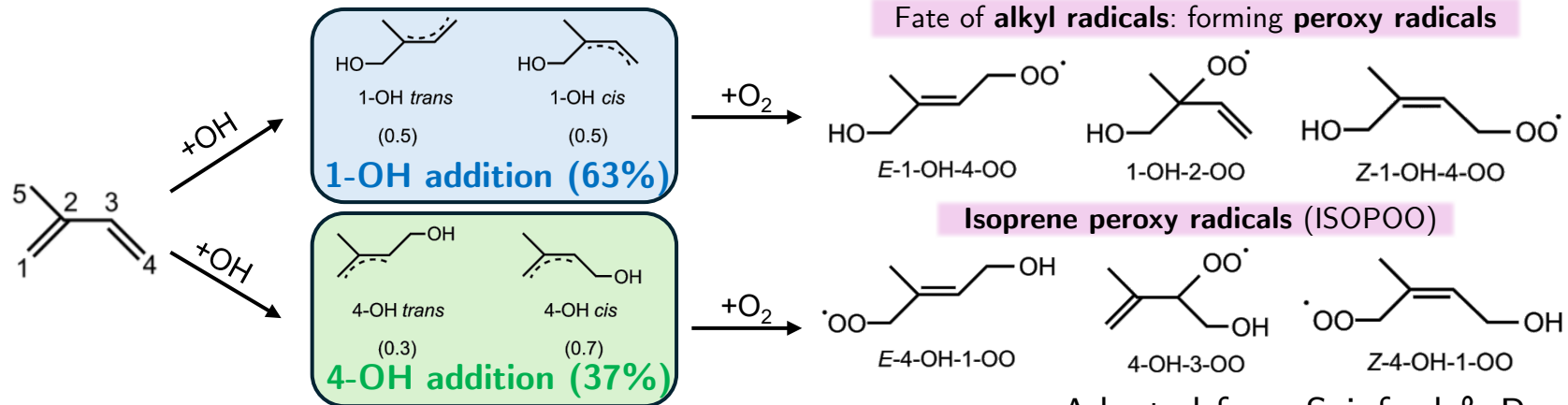
$$\tau_{\text{OH}} = \frac{1}{k_{\text{OH}}[\text{OH}]} = \frac{1}{(2.7 \times 10^{-11}) \exp(390/223) \times 10^6} = 6940 \text{ s} = 1.93 \text{ hr}$$

Standard temperature: 273 K,

$$\tau_{\text{OH}} = \frac{1}{k_{\text{OH}}[\text{OH}]} = \frac{1}{(2.7 \times 10^{-11}) \exp(390/273) \times 10^6} = 8870 \text{ s} = 2.46 \text{ hr}$$

OH is the dominant daytime sink (85%) for isoprene and is faster in the cold upper troposphere.

- OH attacks primarily to the double bond.

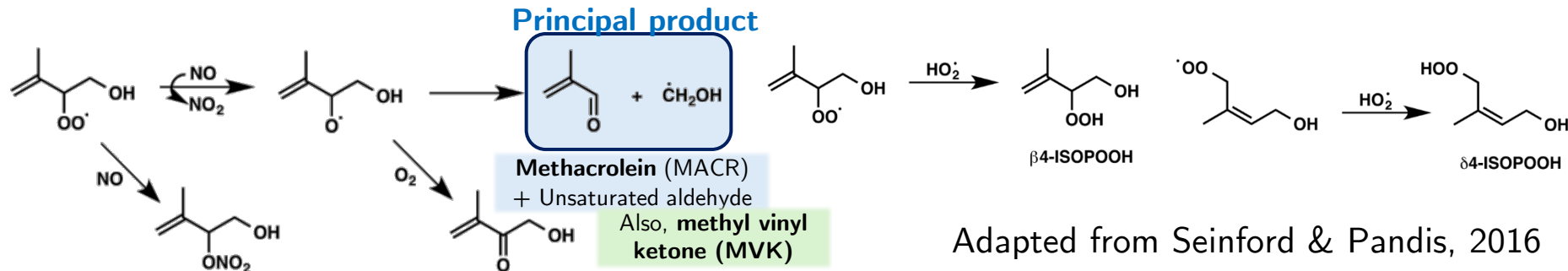


Adapted from Seinfeld & Pandis, 2016

Derivation of Oxidation Products from Isoprene

First-Generation Products

- Fate of the isoprene peroxy radicals depends on the relative concentrations of NO and HO₂. NO₂ may also be important (forming hydroxy peroxy nitrates, IHPN).



Fates of peroxy radicals with NO: stable hydroxy nitrates (IHN) or alkoxy radical, which then decomposes or react with O₂

Adapted from Seinfeld & Pandis, 2016

Fate of peroxy radicals with HO₂: ISOPROOH

Second-Generation Products (Russell et al., 2025)

Initiation Reaction	Products Important for Nucleation Growth
ISOPOOH + OH → C ₅ H ₁₁ O ₄	→ +O ₂ → C ₅ H ₁₁ O ₆ → +HO ₂ → C ₅ H ₁₂ O ₆ → +NO → C ₅ H ₁₁ NO ₇
IHN + OH → C ₅ H ₁₀ NO ₅	→ +O ₂ → C ₅ H ₁₀ NO ₇ → +HO ₂ → C ₅ H ₁₁ NO ₇ → +NO → C ₅ H ₁₀ N ₂ O ₈

The hydroperoxides and nitrates are permanent radical sinks to build particles. Peroxynitrates are temporary reservoirs storing radicals due to the cold. They do not directly form particles, but critically influence the timing of particle formation and fate of RO₂.

Note. Volatility Range Classification: IVOC range ~10²-10⁶ μg m⁻³, SVOC range ~10⁻¹-10² μg m⁻³, ELVOC range <10⁻⁴ μg m⁻³.

Under boundary-layer temperatures,

$C_{\text{ISOPOOH}}^* \sim 5.6 \times 10^4 \mu\text{g m}^{-3}$, $C_{\text{IEPOX}}^* \sim 1.7 \times 10^4 \mu\text{g m}^{-3}$ (**IVOC**)

First generation products are too volatile to drive aerosol formation. Even for upper-tropospheric temperatures, most C₄ products are too volatile to contribute to nucleation or even particle growth (Russell et al., 2025).

Under upper tropospheric conditions (223 K), with the SIMPOL model, $C_{\text{dihydroperoxide}}^* \sim 10^{-7.19} \mu\text{g m}^{-3}$, $C_{\text{mononitrate}}^* \sim 10^{-6.75} \mu\text{g m}^{-3}$, $C_{\text{dinitrate}}^* \sim 10^{-6.75} \mu\text{g m}^{-3}$ (**ELVOC**)

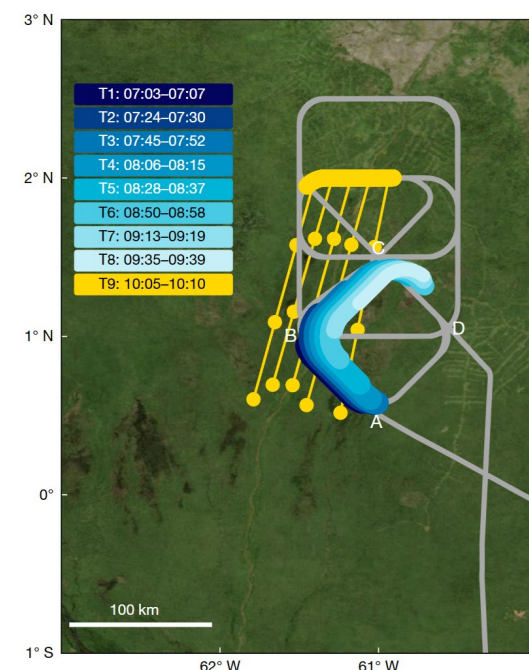
Second-generation IP-OOM are the main drivers of NPF.

In-Situ Measurements of New Particle Formation in the Upper Troposphere

- ▶ Surprisingly **high isoprene mixing ratio of 1 ppbv during nighttime in the upper troposphere** (Shen et al., 2024).
- ▶ IPOOM contributes substantially to SOAs, but the products are **too volatile to nucleate in the boundary layer**. At 223 K, the saturation vapor pressures of these second-generation molecules drop by several orders of magnitude.

Chemistry of the Atmosphere: Field Experiment in Brazil Campaign (CAFE-Brazil, 2022-2023) (Curtius et al., 2024)

- ▶ **Aircraft measurements** over the Amazon above 8 km observed massive particle bursts after 2 h of sunrise.
- ▶ At sunrise, photochemical oxidation forms **isoprene mononitrates & dinitrates**, consistent with the bursts.
- ▶ Freshly formed particles (2-5 nm) frequently exceeded $10,000 \text{ cm}^{-3}$.
- ▶ **Hypothesis**: Monoterpenes play no significant role; ions assist but are not the main driver.
- ▶ **Implications for further investigations**: quantifying the complete gas-phase oxidation cascade from isoprene to dinitrates; trace acids enhancement for nucleation.

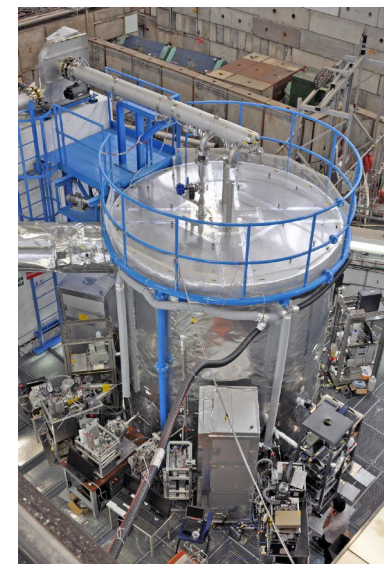
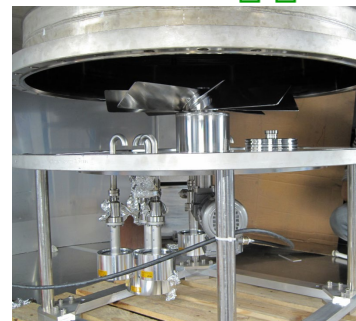
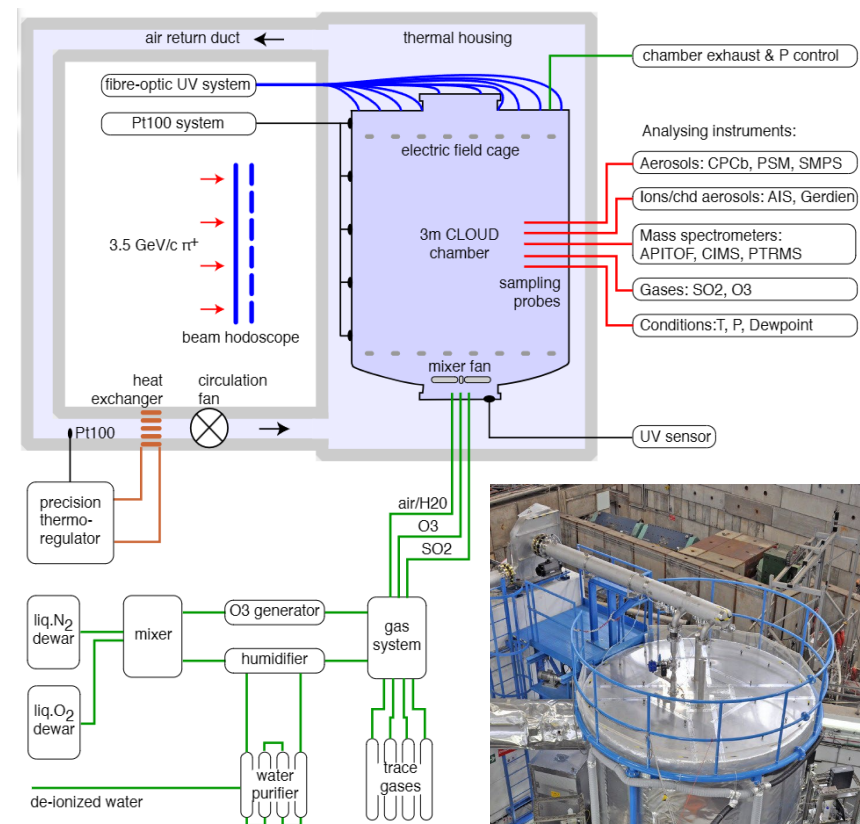


Source: Curtius et al., 2024

CLOUD Experiment (Shen et al., 2024)

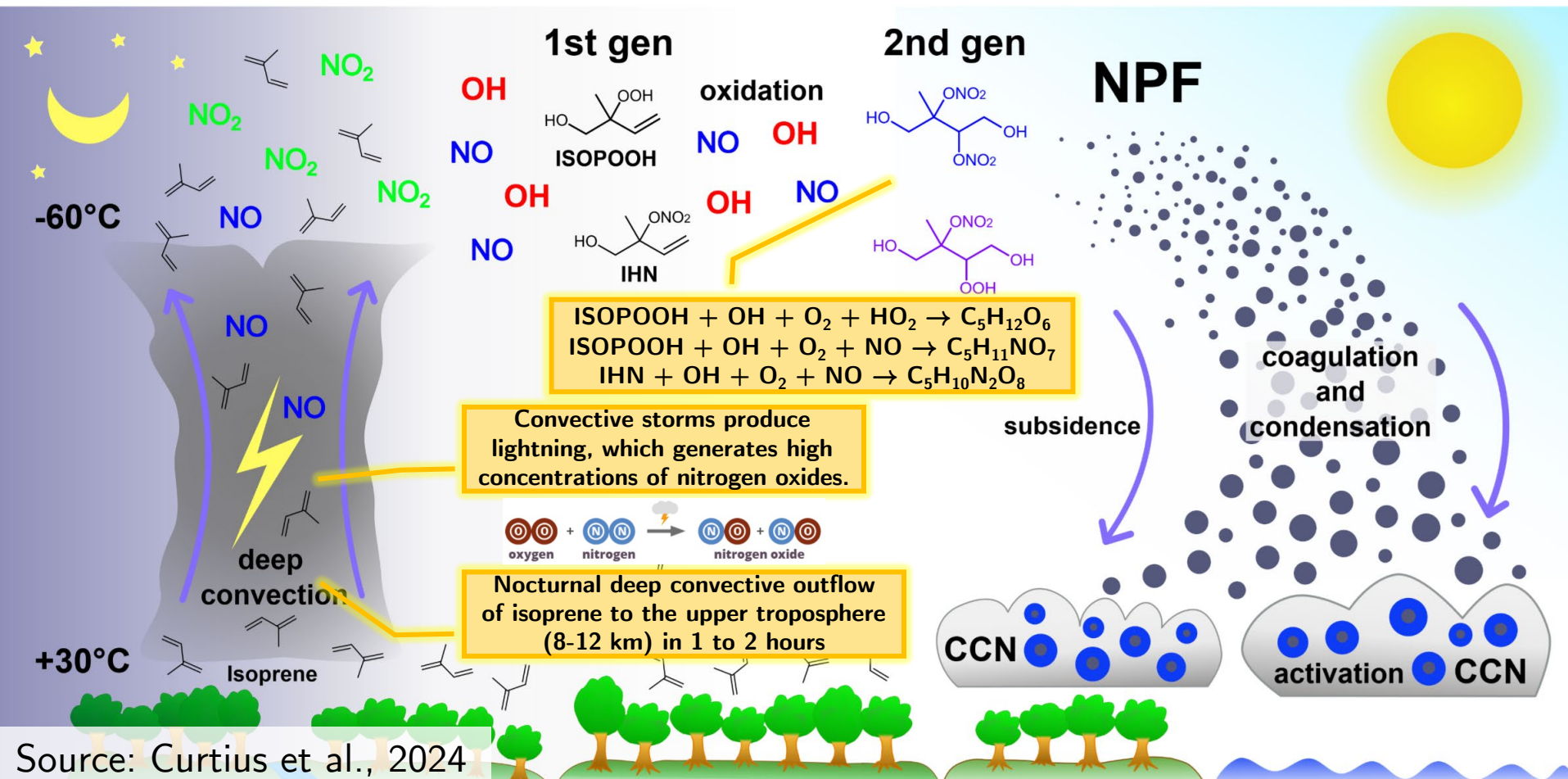
- CLOUD chamber at 223 K, 950 mbar, isoprene+OH+O₂, no monoterpenes, no added acids.
- **Result: IP-OOM alone nucleate** at concentrations found in the UT ($J_{1.7}$ up to $\sim 1 \text{ cm}^{-3} \text{ s}^{-1}$ without acids).
- Adding trace H₂SO₄ or HIO_x ($\geq 10^5 \text{ cm}^{-3}$) enhances nucleation 100-fold (rate of 1.7-nanometer particle formation, $J_{1.7}$, up to $48 \text{ cm}^{-3} \text{ s}^{-1}$).
- Nucleation rate scales with $[\text{IP-OOM}] \times [\text{Acid}]$ - linear product indicates 1:1 acid:IP-OOM dimer is the critical step.

Non-nitrate IP-OOM (C₅H₁₂O₆) are more effective nucleators than nitrate IP-OOM (C₅H₁₁NO₇, C₅H₁₀N₂O₈) at the same concentration – but at UT temperatures, nitrates also have extremely low volatility and contribute more.



CERN CLOUD Chamber
Source: CERN, 2009

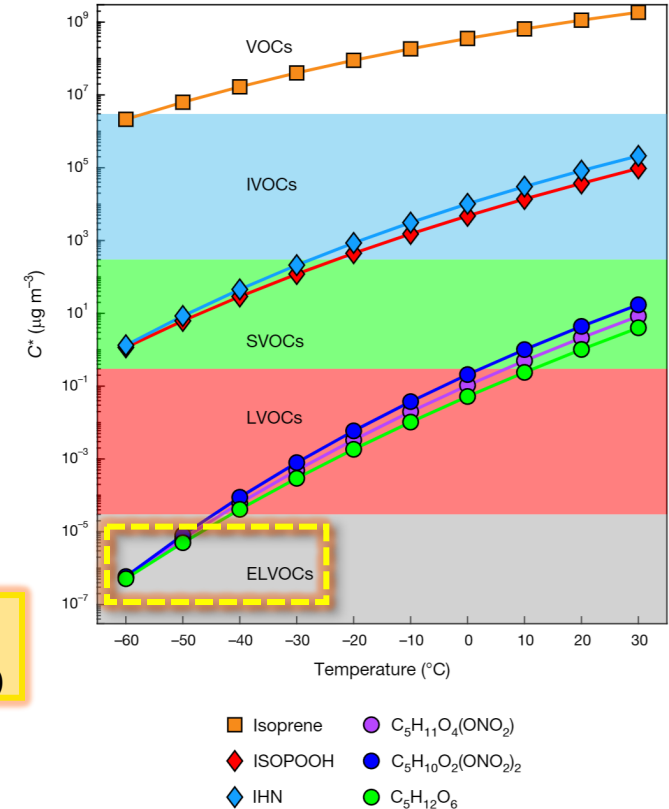
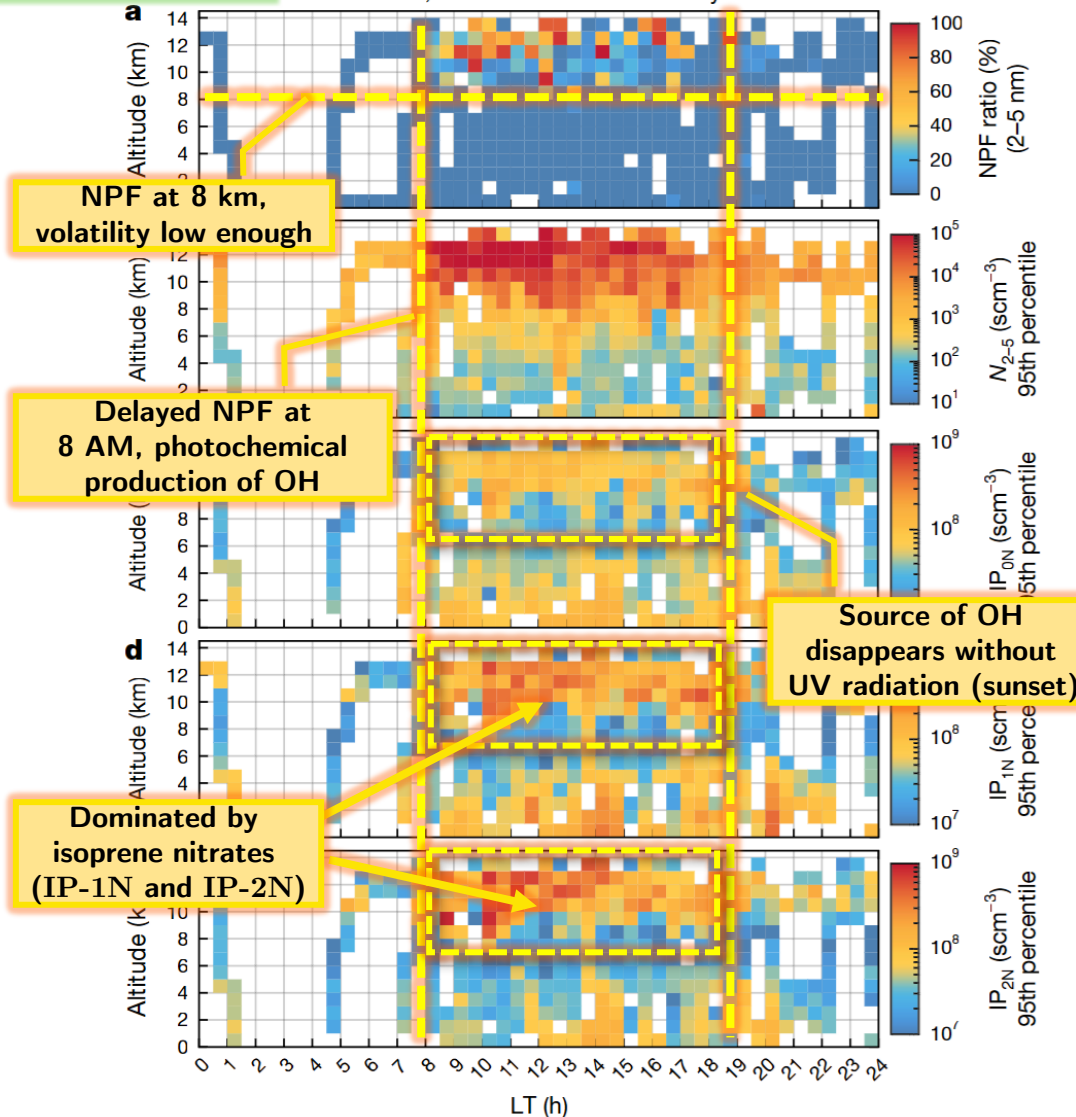
A Full Picture of the NPF Process



Diurnal Cycle of NPF (Curtius et al., 2024)

Flight Measurements

CAFE-Brazil, December 2022 to January 2023

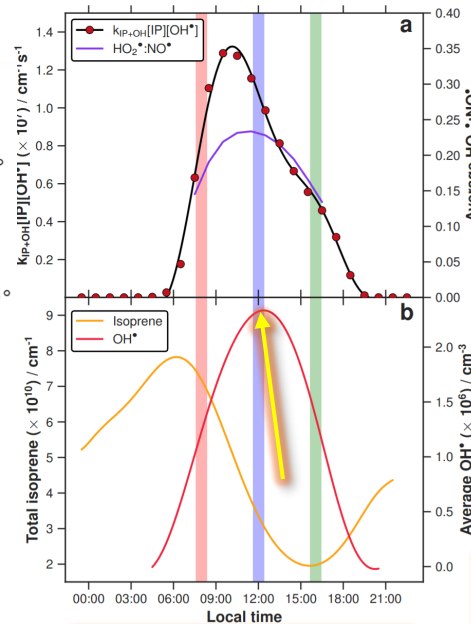
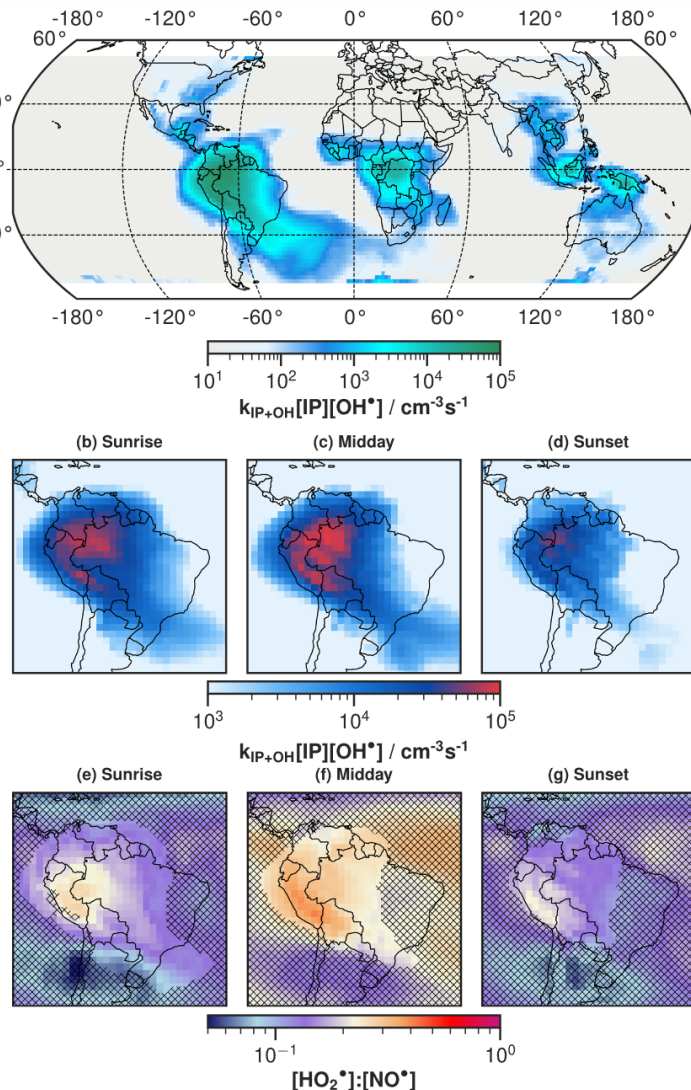


Source: Curtius et al., 2024

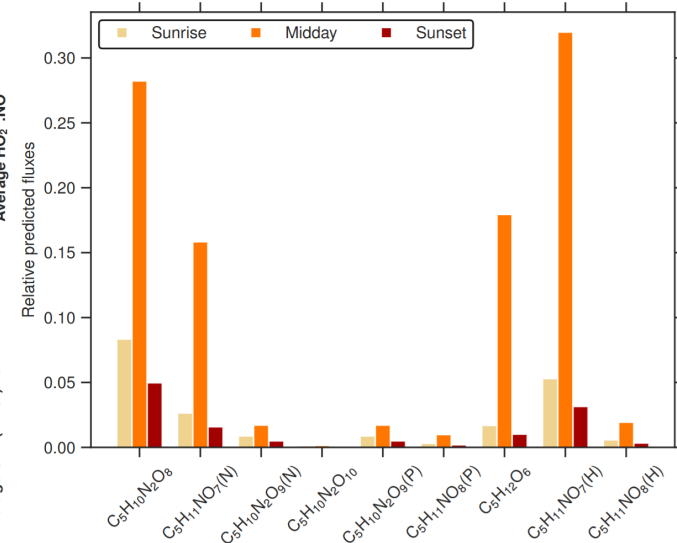
Daily massive particle bursts likely to occur every single morning during wet season across a large portion of the Amazon.

Diurnal Cycle of NPF (Russell et al., 2025)

EMAC model simulation under improved isoprene chemistry mechanisms



HOx peaks at noon
when UV is strongest



Sunrise/Sunset: mononitrates & dinitrates
Midday: Both nitrates & hydroperoxides

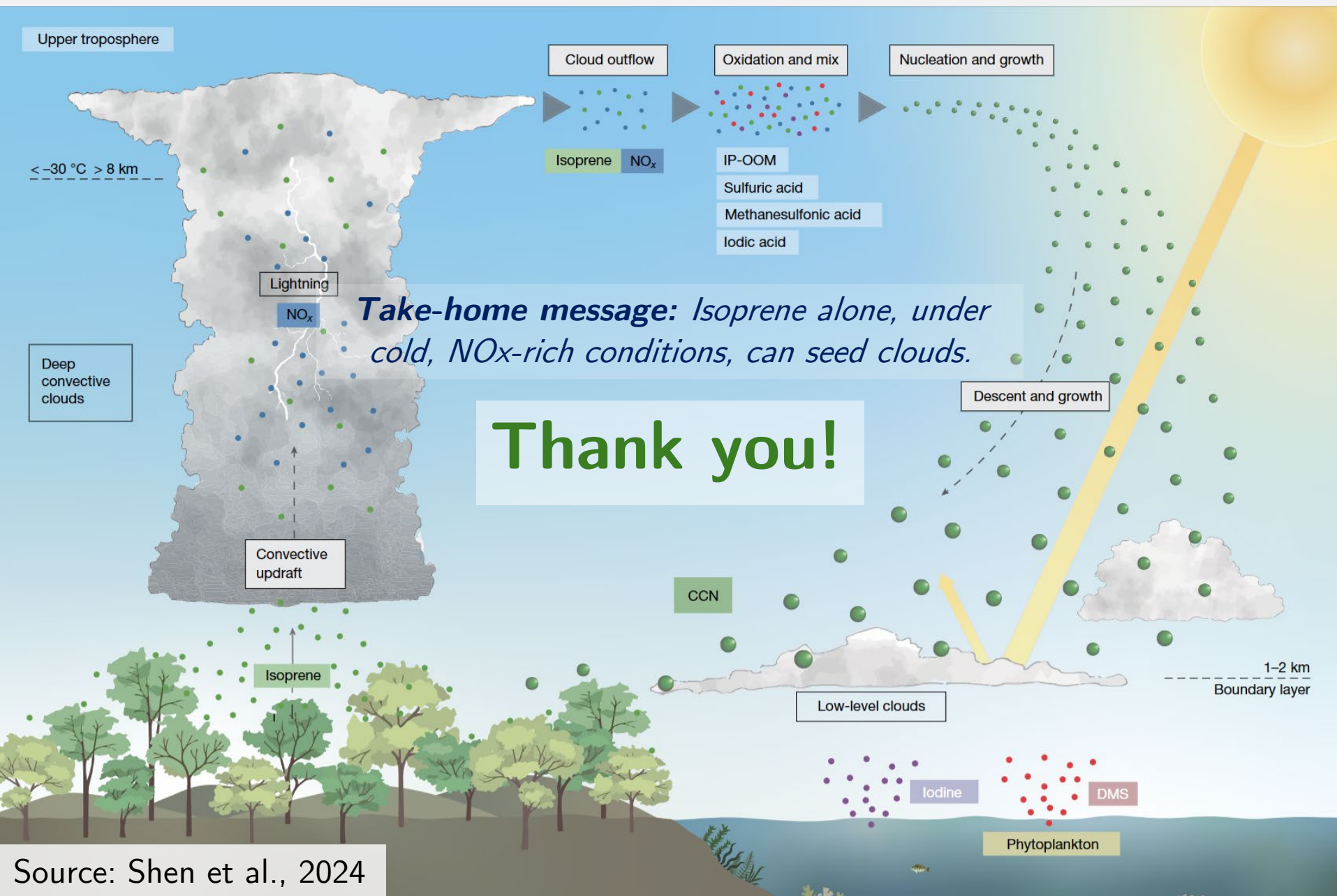
Source: Russell et al., 2025

Early morning (06:00–10:00): isoprene accumulates overnight (slow oxidation by NO_3/O_3), the air is almost entirely IP_{2N} (dinitrate) and IP_{1N} (mononitrate). This is because NO is the dominant reactant early on.

Afternoon (12:00–18:00): As HO_2 increases with the sun, the IP_{0N} (hydroperoxide) fraction grows, but not enough to become dominant due to high NO concentrations.

Implications and Conclusion

- ▶ **7000 mesoscale convective systems** occur over the Amazon each year, typically lasting for 4 hours with high lightning activity, NPF from oxidation of isoprene combined with NO_x represents a notable mechanism in tropical troposphere. New particles in the upper troposphere can be transported **over large distances** well beyond the Amazon region, affecting **CCN abundance at lower altitudes in the tropics** (Curtius et al., 2024; Shen et al., 2024).
- ▶ **Peroxynitrate pathway** should be considered **for isoprene chemistry under upper tropospheric conditions in numerical models** (Russell et al., 2025).
- ▶ IP-OOM yield from reacted precursors is 68% ($\text{IP}_{0\text{N}}$ 36%, $\text{IP}_{1\text{N}}$ 31%, $\text{IP}_{2\text{N}}$ 42%), justifying the high efficiency of particle formation (Russell et al., 2025).
- ▶ **Remaining challenges:**
 - ✓ Accurate low-temperature rate coefficients for $\text{RO}_2 + \text{NO}_2$ and $\text{RO}_2 + \text{HO}_2$ for functionalised RO_2 .
 - ✓ Volatility of specific isomers – FIGAERO and SIMPOL show large discrepancies.
 - ✓ Heterogeneous chemistry of organic nitrates (hydrolysis, uptake) – currently poorly constrained.
 - ✓ Interactions with monoterpenes in the UT – need follow-up experiments.



Source: Shen et al., 2024

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